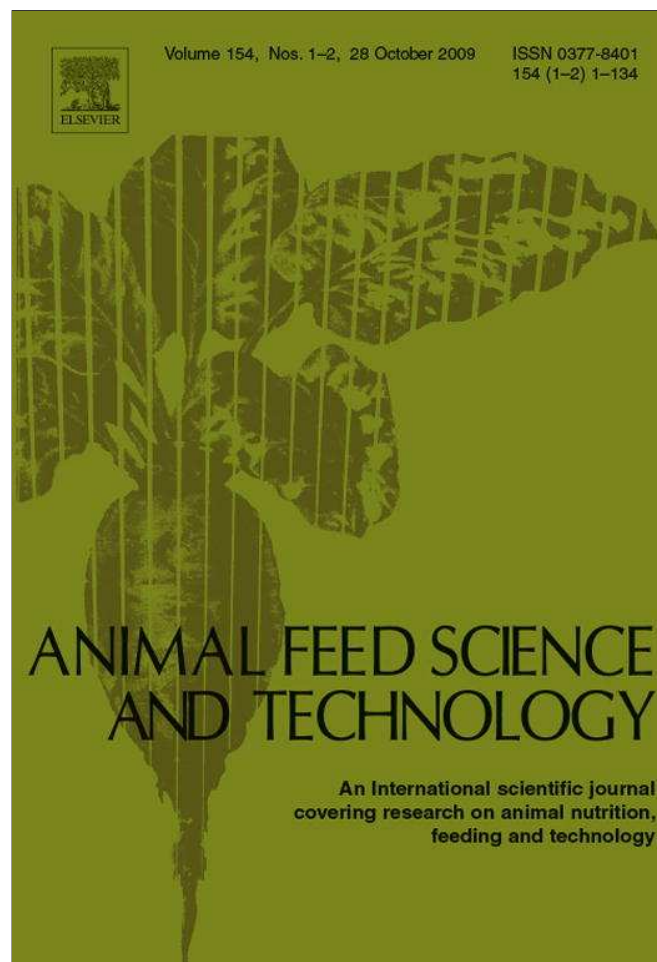




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## Short communication

# Performance, chewing activity, and ruminal parameters in yearling beef steers fed early-harvested sorghum silage: Effect of chop length and wheat straw addition

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## ABSTRACT

Two experiments were conducted to evaluate effects of silage chop length (SCL) and long wheat straw (WS) addition on performance, ruminal parameters, and chewing activity of yearling beef steers. The aim of the first experiment was to measure dry matter (DM) intake, average daily gain (ADG) and feed conversion (FC). For this, 59 Angus yearling steers were assigned to one of the four treatments in a  $2 \times 2$  factorial arrangement. The factors were SCL (*i.e.*, 5.8 and 9.4 mm mean particle length, MPL) and WS addition (*i.e.*, 0 and 100 g/kg). Diets consisted of 792 g/kg of early-harvested sorghum silage (SS, *Sorghum bicolor* (L.) Moench.), 200 g/kg of pelleted sunflower meal, and 8 g/kg of urea and mineral–vitamin supplements. The amount of SS was reduced by 100 g/kg when WS was added to the diet. The second experiment, which aimed to evaluate ruminal parameters and chewing activity, consisted of two treatments with four rumen-cannulated steers fed with a diet comprised of finely chopped SS (5.8 mm MPL) either with or without WS. In the first experiment, SCL had no effect on DM intake, ADG or FC, while addition of WS depressed DM intake and ADG and had a negative effect on FC, without interactions between the factors. In the second experiment, WS addition reduced DM intake and volatile

**Abbreviations:** aNDF, neutral detergent fiber assayed with a heat stable amylase and expressed inclusive of residual ash; ADF, acid detergent fiber expressed inclusive of residual ash; ADG, average daily gain; BW, body weight; CP, crude protein; DM, dry matter; ET, eating time; FC, feed conversion; lignin(pm), lignin determined by oxidation of lignin with permanganate; MPL, mean particle length; PEF, physical effectiveness factor; peNDF, physically effective NDF; PSPS, Penn State Particle Separator; RT, ruminating time; SCL, silage chop length; SS, sorghum silage; TCT, total chewing time; TLC, theoretical length of cut; VFA, volatile fatty acid; WS, wheat straw; WSC, water soluble carbohydrates.

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fatty acid concentration, but did not affect rumen pH or chewing activity. Results suggest that WS addition to these SS-based diets containing a high aNDF did not improve the ruminal environment, did not affect chewing activity and had a negative effect on steer performance, while SCL may not affect performance of beef steers.

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## 1. Introduction

Sorghum silage (SS, *Sorghum bicolor* (L.) Moench.) is an important feedstuff used in meat production worldwide, however animal performance has been highly variable depending on silage quality (Van Amburgh et al., 2004), which is largely determined by grain content and stover digestibility (Di Marco et al., 2002; Miron et al., 2005). Both variables are markedly affected by stage of maturity at harvest (Johnson et al., 1999; Andrae et al., 2001). Since grain content increases, and stover digestibility decreases, with advancing maturity, there is a conflict between grain content and stover quality (Johnson et al., 1999). Early harvest of sorghum may allow higher stover quality, thereby yielding higher neutral detergent fiber (NDF) digestibility (Hanna et al., 1981).

Fine chopping during harvest improves silage packing density and fermentation, and aids in preserving nutritive value (McDonald, 1981). However, feedstuffs with small particle size may have adverse effects on digestion. For example, ration physical effectiveness may be reduced, thereby decreasing mastication, salivary buffer secretion, ruminal pH (Mertens, 1997; Allen, 1997), cellulolytic activity and fiber degradation (Mould and Ørskov, 1983; Hoover, 1986; Roger et al., 1990). It may increase the rate of passage, thereby reducing silage ruminal retention time (Rodrigue and Allen, 1960; Pearce and Moir, 1964; Jaster and Murphy, 1983; Varga and Kolver, 1997), with a consequent reduction in dry matter (DM) digestibility (Miller et al., 1969; Waldo and Smith, 1972; Allen and Mertens, 1988). Physical effectiveness of silage-based rations can be improved by increasing forage chop length or by addition of long fiber to the diet (Beauchemin et al., 1994; Couderc et al., 2006). Long fiber addition increases mastication and improves ruminal environment, and may enlarge the dorsal floating 'raft' in the rumen dorsal sac. This raft may entrap small particles through mechanical entanglement, thereby increasing ruminal retention time of particles that would otherwise sink and pass to the omasum (Welch, 1986; Sutherland, 1988; Zebelli et al., 2007). A great number of studies have been conducted in the last several years (Beauchemin et al., 2003; Beauchemin and Yang, 2005; Couderc et al., 2006; Zebelli et al., 2007) to determine effects of forage particle size and physical effectiveness of the ration on DM intake, chewing activity, and the ruminal environment in high producing dairy cows. However, results are not conclusive (Beauchemin and Yang, 2005; Zebelli et al., 2007) and research is scarce to determine if results obtained in dairy cattle are applicable to beef cattle.

This study was completed with two objectives. First, to evaluate effects of SCL and WS addition on performance of yearling beef steers fed a diet comprised of approximately 0.80 early-harvested SS. Secondly, to determine effects of WS addition on ruminal parameters (i.e., pH, volatile fatty acids (VFA) and  $\text{NH}_3$ ) and chewing activity.

## 2. Materials and methods

### 2.1. Experiments

Two experiments were conducted at the Balcarce Experimental Station of the National Institute of Agriculture Technology (INTA), in Argentina (37°45'lat. S, 58°18'long. W). The first was a 77-day production study and the second was a 28-day digestion study. The sorghum crop (Dairy Master®, Druetto, Argentina) was sown on December 1st 2005, harvested at the half-bloom stage (Vanderlip and Reeves, 1972) on March 20th 2006 and chopped at two theoretical lengths of cut (TLC), being 6 mm (fine) and 23 mm (coarse), with a forage harvester (model FX 375, New Holland, New Holland, PA, USA). The chopped forage was packed in stack silos and covered with black plastic for approximately 3 mo. Wheat straw (WS) bales (50 kg) were purchased

from a private farm and chopped with a commercial forage chopper to approximately 50 mm lengths.

## 2.2. Experiment 1

Angus yearling steers ( $n = 59$ ) weighing  $179 \pm 13.7$  kg body weight (BW) were blocked by initial BW and assigned randomly within blocks to four treatments in a  $2 \times 2$  factorial arrangement. The factors were: chop length of SS (6 mm versus 23 mm TLC) and two levels of WS addition (0 g/kg versus 100 g/kg, DM basis). Diets consisted of 792 g/kg of early-harvested SS, 200 g/kg of pelleted sunflower meal, and 8 g/kg of urea and mineral–vitamin supplements [composition: urea 530 g/kg, Cl 98 g/kg, Na 65 g/kg, Ca 199 g/kg, Mg 18 g/kg, Mn 2.6 g/kg, Zn 2.7 g/kg, Cu 1.1 g/kg, Se 8.6 ppm, I 25 ppm, Co 5.4 ppm]. In treatments with addition of WS, the amount of SS was proportionally reduced.

Steers were fed once a day at 0700 h in 12 open-air corrals (4 m  $\times$  5 m, dirt floor) with five animals each (*i.e.*, three corrals/treatment), with the exception of one corral in which one animal was removed before the beginning of the study. Rations were mixed and administered by means of a mixer (Mainero, 2910, Córdoba, Argentina) and WS was added to the top of the bunks and hand-mixed daily. Orts were collected and weighed on three consecutive days each week (*i.e.*, Tuesday, Wednesday, Thursday) during the whole study and DM intake of each corral was calculated by difference between the weight of offered feed minus the weight of orts adjusted for their respective DM contents. The DM content of orts was assessed by oven-drying at 60 °C for 48 h.

Average daily gain (ADG) was estimated through simple regression analysis of BW data obtained from 6 fortnightly unshrunk weight measurements. Steers were weighed at 0800 h. Initial and final measurements were in duplicate on consecutive days to reduce daily variations due to digestive tract fill (Stock et al., 1983). Feed conversion (FC) was calculated (kg DM intake/kg ADG) from DM intake and ADG data.

Feed samples were collected monthly and pooled for laboratory analysis. Composite samples were oven dried at 60 °C for 48 h for DM content determination and ground to pass a 1 mm screen in a Wiley mill (Thomas-Wiley, Philadelphia, PA, USA). The OM content was calculated by difference between DM and ash content (550 °C overnight). Chemical composition was determined as follows: aNDF, ADF and lignin(pm) content as described by Van Soest et al. (1991), with aNDF determined with a heat stable amylase and expressed inclusive of residual ash, while lignin(pm) was determined by oxidation with permanganate. Starch was determined as described by McRae and Armstrong (1968), water soluble carbohydrates (WSC) as described by Bailey (1958) and N by Kjeldahl (method 990.02, AOAC, 1990). Mean particle length (MPL) of SS was assessed using a Nasco (Fort Atkinson, WI, USA) Penn State Forage Particle Separator (PSPS) (Heinrichs, 1996; Lammers et al., 1996). This sieving device is comprised of two screens of 19 and 8 mm (top and middle screen, respectively) and a solid bottom pan. Two samples (750 g, as fed) of SS were placed on the top screen and the PSPS was shaken 40 times, as described by Heinrichs (1996). The MPL of silage and mixed ration was estimated by graphing the distribution of forage particles on Weibull paper as described by Lammers et al. (1996). Physically effective NDF was obtained as the proportion of particles longer than 8 mm (physical effectiveness factor, PEF) multiplied by dietary aNDF content (Lammers et al., 1996; Beauchemin et al., 2003).

The DM intake data were analyzed as repeated measures with the MIXED procedure of SAS (SAS Institute, 2001), while ADG and FC data were analyzed using the GLM procedure of SAS (SAS Institute, 2001). Differences among means with  $P < 0.05$  were accepted as representing statistically significant differences, while differences among means with  $0.05 < P < 0.10$  were accepted as representing tendencies to difference.

## 2.3. Experiment 2

Four Angus  $\times$  Hereford crossbred yearling steers with permanent ruminal cannulas were housed individually, in open-air corrals, and fed once daily at 0700 h the finely chopped early-harvested SS (5.8 mm MPL) with either 0 or 100 g/kg WS. They were assigned randomly to sequence in a crossover design balanced for carryover effects. Period lengths were 14 days, with 12 days for adaptation to treatments and 2 days for data and sample collection. To accelerate ruminal adaptation, the reticulo-

ruminal contents were removed and interchanged between steers before the beginning of the second period (Koenig et al., 2003).

Orts were collected daily and individual DM intake was obtained by difference between feed offered and refused, adjusting for the DM content of the Orts.

Chewing activity was visually determined by observation for 24 h on day 13 of each period. Observed behavior was recorded every 5 min and classified as eating, ruminating or idling (*i.e.*, neither eating nor ruminating, which included resting, drinking, licking and grooming). Each activity was assumed to persist for the entire 5 min interval to estimate eating time (ET; min/day) and ruminating time (RT; min/day), while idle time was not recalculated. Total chewing time (TCT; min/day) was calculated as the sum of ET and RT. Eating, ruminating and total chewing time per kilogram of DM intake (min/kg of DM intake) and per kilogram of aNDF intake (min/kg of aNDF intake) was estimated using average DM intake measured in that period since it was assumed that chewing activity reflected DM intake of the previous days.

On day 14 of each period, rumen digesta samples were collected every 4 h (*i.e.*, 0700, 1100, 1500, 1900, 2300, 0300 h) from the dorsal, ventral and caudal areas of the rumen and squeezed through a four-layer cheesecloth. The pH was measured in filtered ruminal fluid, immediately after collection, using a hand-held pH electrode (Cole-Palmer®, Chicago, IL, USA). A 20 ml aliquot of filtered fluid was then placed into plastic pots containing 4 ml of 190 g/l metaphosphoric acid and 4 ml of 64 g/l sulphuric acid and stored at  $-20^{\circ}\text{C}$  for later analysis. Subsequently, samples were thawed and centrifuged at  $10,000 \times g$  for 10 min at  $4^{\circ}\text{C}$  to obtain a clear supernatant, which was then analyzed for  $\text{NH}_3\text{-N}$  using a Kjeltac 1030 analyzer (Tecator, Hoganas, Sweden) and for VFA using gas chromatography (Varga and Prigge, 1982).

The DM intake data were analyzed using the GLM procedure of SAS, while VFA, pH,  $\text{NH}_3$  were analyzed with the MIXED procedure of SAS (SAS Institute, 2001). Chewing activity data were analyzed by the MIXED procedure of SAS, according to the model:

$$Y_{ijkl} = \mu + P_i + A_j + S_k + T_l$$

where  $Y_{ijkl}$  is the dependent variable,  $\mu$  is the population mean,  $P_i$  is the period effect ( $i = 1, 2$ ),  $A_j$  is the random effect of the animal ( $j = 1-4$ ),  $S_k$  is the sequence effect ( $k = 1, 2$ ) and  $T_l$  is treatment effect ( $j = 1, 2$ ). Differences among means with  $P < 0.05$  were accepted as representing statistically significant differences, while differences among means with  $0.05 < P < 0.10$  were accepted as representing tendencies to differences.

### 3. Results

#### 3.1. Feedstuff characterization

There were no differences in chemical composition between finely and coarsely chopped silage, which had 643 g/kg DM of aNDF and 85 g/kg DM of lignin(pm), and low starch and water soluble carbohydrate (WSC) contents (Table 1). The WS used was very poor quality and contained high aNDF levels. The pelleted sunflower meal had a CP content of 330 g/kg, which is within the range expected for this kind of feedstuffs (Villamide and San Juan, 1998).

Rations were formulated to achieve similar CP and metabolizable energy (ME, MJ/kg) levels (Table 2). Differences in ADF (+1.15%) content were caused by the addition of WS. Varying the TLC of silage increased its MPL by 3.6 mm and affected ration MPL (+3.2 mm) more markedly than WS addition (+2.05 mm) (Table 2). Even though ration MPL was affected by WS ( $P < 0.001$ ), neither the PEF nor the peNDF contents were affected by WS. In contrast, SCL affected the peNDF content of the rations.

#### 3.2. Experiment 1

As there were no interactions between SCL and WS, only main effect means are in Table 3. Animal performance parameters (ADG, FC and DM intake) were not affected by SCL, while WS addition

**Table 1**Feedstuffs ( $n = 4$ ) and mean particle length of sorghum silage and wheat straw as assessed by the Penn State Particle Separator.

| Chemical component                | Sorghum silage |                | Effect $P = ^a$ SCL | Wheat straw     | Sunflower meal |
|-----------------------------------|----------------|----------------|---------------------|-----------------|----------------|
|                                   | Fine           | Coarse         |                     |                 |                |
| Dry matter (g/kg as fed)          | 250 $\pm$ 4.6  | 238 $\pm$ 7.1  | 0.25                | 899 $\pm$ 3.7   | 913 $\pm$ 3.7  |
| DM basis                          |                |                |                     |                 |                |
| CP (g/kg DM)                      | 101 $\pm$ 4.8  | 100 $\pm$ 4.6  | 0.71                | ND <sup>b</sup> | 330 $\pm$ 9.7  |
| Neutral detergent fiber (g/kg DM) | 643 $\pm$ 13.7 | 643 $\pm$ 17.7 | 0.79                | 806 $\pm$ 21.3  | 339 $\pm$ 11.6 |
| Acid detergent fiber (g/kg DM)    | 476 $\pm$ 7.1  | 471 $\pm$ 4.8  | 0.81                | 598 $\pm$ 8.0   | 291 $\pm$ 4.4  |
| Lignin(pm) <sup>c</sup> (g/kg DM) | 85 $\pm$ 1.5   | 89 $\pm$ 3.0   | 0.62                | ND              | ND             |
| Starch (g/kg DM)                  | 8 $\pm$ 3.8    | 14 $\pm$ 4.5   | 0.07                | ND              | ND             |
| WSC <sup>d</sup> (g/kg DM)        | 13 $\pm$ 1.9   | 15 $\pm$ 2.2   | 0.69                | ND              | ND             |
| NH <sub>3</sub> (g/kg total N)    | 150 $\pm$ 7.5  | 66 $\pm$ 2.3   | <0.01               | ND              | ND             |
| MPL <sup>e</sup> (mm)             | 5.8 $\pm$ 0.42 | 9.4 $\pm$ 0.61 | <0.01               | 58 $\pm$ 3.2    | ND             |

<sup>a</sup> Silage chop length.<sup>b</sup> Not determined.<sup>c</sup> Lignin determined by oxidation of lignin with permanganate.<sup>d</sup> Water soluble carbohydrates.<sup>e</sup> Mean particle length.**Table 2**Ingredients and chemical composition, energy content and mean particle length of experimental diets ( $n = 4$ ).

|                               | Fine SS <sup>a</sup> |             | Coarse SS <sup>b</sup> |             | SEM   | P     |       |                 |
|-------------------------------|----------------------|-------------|------------------------|-------------|-------|-------|-------|-----------------|
|                               | 0 g/kg WS            | 100 g/kg WS | 0 g/kg WS              | 100 g/kg WS |       | SCL   | WS    | SCL $\times$ WS |
| Composition (g/kg DM)         |                      |             |                        |             |       |       |       |                 |
| Sorghum silage                | 794                  | 692         | 794                    | 692         |       |       |       |                 |
| Sunflower meal                | 200                  | 200         | 200                    | 200         |       |       |       |                 |
| Urea–mineral mix <sup>c</sup> | 6                    | 8           | 6                      | 8           |       |       |       |                 |
| Wheat straw                   | 0                    | 100         | 0                      | 100         |       |       |       |                 |
| Analyses <sup>d</sup>         |                      |             |                        |             |       |       |       |                 |
| DM (g/kg as fed)              | 380                  | 445         | 371                    | 437         | 12.8  | 0.59  | <0.01 | 0.97            |
| CP (g/kg DM)                  | 163.6                | 161.1       | 163.1                  | 160.6       | 3.30  | 0.71  | 0.27  | 0.85            |
| NDF (g/kg DM)                 | 578.1                | 593.1       | 578.6                  | 593.6       | 11.87 | 0.98  | 0.23  | 1.00            |
| ADF (g/kg DM)                 | 436.1                | 447.4       | 432.7                  | 444.4       | 4.55  | 0.47  | 0.03  | 0.96            |
| ME (MJ/kg DM)                 | 8.74                 | 8.61        | 8.23                   | 8.15        | 0.301 | 0.13  | 0.71  | 0.91            |
| MPL <sup>e</sup> (mm)         | 5.43                 | 7.57        | 8.75                   | 10.72       | 0.456 | <0.01 | <0.01 | 0.86            |
| PEF <sup>f</sup>              | 0.27                 | 0.30        | 0.61                   | 0.62        | 0.022 | <0.01 | 0.38  | 0.64            |
| peNDF <sup>g</sup> (g/kg DM)  | 158                  | 179         | 355.5                  | 370.5       | 13.16 | <0.01 | 0.19  | 0.78            |

<sup>a</sup> Finely chopped sorghum silage.<sup>b</sup> Coarsely chopped sorghum silage.<sup>c</sup> Urea–mineral mix composition: urea 530 g/kg, Cl 98 g/kg, Na 65 g/kg, Ca 199 g/kg, Mg 18 g/kg, Mn 2.6 g/kg, Zn 2.7 g/kg, Cu 1.1 g/kg, Se 8.6 ppm, I 25 ppm, Co 5.4 ppm.<sup>d</sup> Calculated from composition of individual ingredients.<sup>e</sup> Mean particle length.<sup>f</sup> Physical effectiveness factor, determined as the proportion of particles longer than 8 mm.<sup>g</sup> Physically effective NDF, estimated as aNDF content multiplied by its PEF.

decreased ADG and final weight of animals ( $P < 0.05$ ) and tended to depress DM intake (5.43 *versus* 5.13,  $P = 0.07$ ). Accordingly, FC was impaired when WS was included in the ration.

### 3.3. Experiment 2

Mean ruminal pH was not affected by treatments and it never fell below 6.0 (Table 4). However, addition of WS decreased total VFA concentration (100.72 mM/dl *versus* 91.56 mM/dl,  $P = 0.004$ ) as well as acetate, propionate and butyrate concentrations, without affecting the A:P ratio and the NH<sub>3</sub>-N concentration. The ET, RT and TCT were not affected by WS addition.

**Table 3**

Effects of diets based on finely or coarsely chopped early-harvested sorghum silage with and without the addition of wheat straw on performance of yearling beef steers.

|                                    | Silage chop length (SCL) |                     | Wheat straw (WS) |          | SEM   | Effect |      |                       |
|------------------------------------|--------------------------|---------------------|------------------|----------|-------|--------|------|-----------------------|
|                                    | Fine <sup>a</sup>        | Coarse <sup>b</sup> | 0 g/kg           | 100 g/kg |       | SCL    | WS   | SCL × WS <sup>c</sup> |
| DM intake (kg/day)                 | 5.39                     | 5.18                | 5.43             | 5.13     | 0.035 | 0.19   | 0.07 | 0.99                  |
| DM intake (g/kg BW)                | 25.7                     | 24.8                | 25.4             | 25.0     | 0.36  | 0.13   | 0.43 | 1.00                  |
| Initial weight (kg)                | 179.2                    | 178.5               | 180.3            | 177.4    | 1.26  | 0.70   | 0.15 | 0.46                  |
| Final weight (kg)                  | 237.8                    | 237.2               | 243.8            | 231.2    | 2.98  | 0.89   | 0.02 | 0.43                  |
| ADG <sup>d</sup> (kg/day)          | 0.71                     | 0.73                | 0.78             | 0.66     | 0.027 | 0.69   | 0.02 | 0.24                  |
| FC <sup>e</sup> (DM intake/kg ADG) | 7.61                     | 7.22                | 6.99             | 7.85     | 0.217 | 0.25   | 0.03 | 0.14                  |

<sup>a</sup> Finely chopped sorghum silage (5.8 mm MPL).

<sup>b</sup> Coarsely chopped sorghum silage (9.4 mm MPL).

<sup>c</sup> Interaction between treatment factors.

<sup>d</sup> Average daily gain.

<sup>e</sup> Feed conversion.

**Table 4**

Effect of the addition of wheat straw to early-harvested sorghum silage-based diets on ruminal pH, NH<sub>3</sub>, VFA concentrations and chewing activity parameters.

|                             | Treatments |             | SEM   | P     |
|-----------------------------|------------|-------------|-------|-------|
|                             | 0 g/kg WS  | 100 g/kg WS |       |       |
| Ruminal parameters          |            |             |       |       |
| Mean ruminal pH             | 6.74       | 6.82        | 0.123 | 0.22  |
| NH <sub>3</sub> (mg/dl)     | 19.71      | 20.94       | 2.177 | 0.40  |
| Total VFA (mMol/dl)         | 100.72     | 91.56       | 2.072 | <0.01 |
| Acetate (mMol/dl)           | 70.73      | 64.24       | 1.548 | <0.01 |
| Propionate (mMol/dl)        | 18.81      | 17.4        | 0.34  | <0.01 |
| A:P ratio <sup>a</sup>      | 3.78       | 3.73        | 0.063 | 0.56  |
| Butyrate (mMol/dl)          | 7.24       | 6.41        | 0.229 | <0.01 |
| Chewing activity parameters |            |             |       |       |
| Eating time                 |            |             |       |       |
| min/day                     | 336        | 331         | 7.6   | 0.66  |
| min/kg of DM                | 62         | 68          | 2.4   | 0.16  |
| min/kg of NDF               | 107        | 114         | 4.1   | 0.28  |
| Ruminating time             |            |             |       |       |
| min/day                     | 481        | 472         | 39.2  | 0.85  |
| min/kg of DM                | 88         | 97          | 5.7   | 0.35  |
| min/kg of NDF               | 152        | 163         | 9.7   | 0.48  |
| Total chewing time          |            |             |       |       |
| min/day                     | 818        | 804         | 43.2  | 0.82  |
| min/kg of DM                | 150        | 165         | 6.2   | 0.17  |
| min/kg of NDF               | 259        | 277         | 10.6  | 0.29  |

<sup>a</sup> Acetate:propionate ratio.

## 4. Discussion

Varying the TLC of silage increased ration MPL more markedly than WS addition, probably because MPL was estimated from wet feeds, considering the proportion of each ingredient on an as fed basis, as indicated by Lammers et al. (1996) and Heinrichs (1996). Therefore, although the WS proportion represented 100 g/kg on a DM basis, it only represented 41.5 g/kg of the as fed ration, thus having a low impact on MPL.

In accordance with the early stage of plant development at harvest, DM content of both silages was low (Johnson et al., 1999; Andrae et al., 2001). Furthermore, starch was almost negligible, while aNDF, ADF and lignin(pm) contents were high. The content of WSC was also negligible, which

may suggest certain degree of leaching due to the high moisture content of the crop at harvest (Castle and Watson, 1973). Therefore, leaching may explain the high contents of aNDF, ADF and lignin(pm).

The lack of effect of SCL on DM intake is consistent with numerous studies reporting a variable response to silage particle size. While some researchers observed an increased DM intake in response to smaller SCL (Wilkinson et al., 1978; Schwab et al., 2002; Kononoff et al., 2003; Couderc et al., 2006), others did not report any effect (Bal et al., 2000; Beauchemin and Yang, 2005). The absence of a difference in DM intake is consistent with the lack of effect on ADG and FC. Since these variables depend on ME intake, and this in turn relies on DM intake and its digestibility, the absence of differences in ADG and FC and that DM intake did not differ between treatments suggests that there were no differences in DM digestibility.

The effect of WS addition on DM intake may be because in high forage diets intake capacity is limited due to restricted flow of digesta from the forestomachs (Allen, 1997). Tjardes et al. (2002) found that addition of inert bulk into the rumen of steers fed high NDF silage-based rations depressed DM intake and reduced ruminal NDF turnover. Ørskov et al. (1991) found a DM intake depression when poor quality WS was added to beef steers diet. Since the WS used in our experiments had high aNDF content (i.e., 800 g/kg DM), it increased the filling property of the diet (Mertens and Ely, 1982; Mertens, 1987), which is related to NDF content (Allen, 1997) as well as to NDF digestibility (Conrad et al., 1964). Furthermore, hay or straw may have a longer ruminal retention time than silage (Wattiaux et al., 1992).

The peNDF content of diets can be estimated by means of two different techniques: (i) considering the proportion of DM longer than 8 mm in the PSPS screens (peNDF<sub>>8</sub>, Lammers et al., 1996), or (ii) considering the proportion of DM longer than 1.18 mm in a dry sieving technique (peNDF<sub>>1.18</sub>, Mertens, 1997). Results of these techniques differ widely and values cannot be used interchangeably (Zebelli et al., 2007). Approximately 0.11 peNDF<sub>>8</sub> would be adequate to maintain ruminal pH above 6.0 (Mertens, 1997; Beauchemin et al., 2003), considered the minimum pH for adequate fiber digestion (Stewart, 1977; Mould and Ørskov, 1983; Mouriño et al., 2001). In the present work, the ruminal environment of both treatments would be considered to be within the optimum range for fiber degradation, and pH values never dropped below 6.0. This is consistent with the peNDF<sub>>8</sub> contents of diets which, despite the fineness of SS chopping, were above the minimum recommended level. Ruminal pH increases quadratically with increasing intakes of peNDF, reaching an asymptotic plateau at values above 6.2 (Mertens, 1997; Zebelli et al., 2007). The lack of differences in pH between treatments may reflect that diets were above the level of peNDF in which ruminal pH is affected by peNDF intake. According to Zebelli et al. (2007) and Beauchemin et al. (2003) considerations, peNDF<sub>>8</sub> intakes above 0.15 DM would not produce any further increase in ruminal pH. However, the concept of peNDF does not consider differences in degradability and acidogenic potential of diets (Zebelli et al., 2007). As stated by Beauchemin and Rode (1997), peNDF needs to depend on the amount of highly fermentable components in the ration and on ruminal degradability of these fractions. Steers fed diets like the ones used in our experiments (i.e., low in non-structural carbohydrates) might have low peNDF requirements to maintain an adequate ruminal pH for cellulolytic activity (Beauchemin and Rode, 1997). High NDF content of diets may compensate for low intakes of peNDF (Beauchemin and Buchanan-Smith, 1989; Beauchemin et al., 1994; Couderc et al., 2006; Zebelli et al., 2007) and addition of coarsely processed forages would not be necessary to maintain adequate ruminal pH, or a high chewing activity, regardless of the fineness of the silage chopping.

Kononoff et al. (2003) have stated that differences in the A:P ratio depend on WSC and starch intake. Our diets did not differ in WSC and starch content, and TCT was similar for both treatments, thus explaining the lack of difference in the A:P ratio.

Chewing activity variables were high, similar to those reported by Deswysen et al. (1987) and Luginbuhl et al. (2000), who analyzed chewing activity in beef cattle at levels of intake below 0.02 of BW, as well as to those reported by Teller et al. (1989), who reported chewing behavior in non-lactating Fresian cows at 0.02 BW DM intake. The high chewing activity per kilogram of DM intake is because the steers were light-weight (De Boever et al., 1990) with low DM intake (Sudweeks et al., 1980).

## 5. Conclusions

In sorghum silage-based diets, with a high content of aNDF, silage chop length did not affect performance, whereas addition of poor quality coarsely processed wheat straw impaired steer performance by reducing dry matter intake, average daily gain and feed conversion. Ruminal parameters indicate that, in spite of the fineness of chopping, physical effectiveness was adequate to maintain ruminal pH within the range desired for a high cellulolysis, because diet NDF content may have contributed to keeping chewing activity at a high level.

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